

The Solow Model

1 The Stylized Facts about Growth

The modern study of economic growth starts in the post-war period and was mostly motivated by the experience of the developed world. In his classical article (Kaldor 1957), Nicolas Kaldor stated some basic facts that he observed economic growth seemed to satisfy, at least in those countries. These came to be known as the Kaldor facts, and the main challenge of growth theory as initially constituted was to account simultaneously for all these facts. But, what were these Kaldor facts? Here they are:

- i. Output per worker shows continuous growth, with no tendency to fall.
- ii. The capital-output ratio is nearly constant. (But what is capital?)
- iii. Capital per worker shows continuous growth (... follows from the other two).
- iv. The rate of return on capital is nearly constant (real interest rates are flat).
- v. Labour and capital receive constant shares of total income.
- vi. The growth rate of output per worker differs substantially across countries (and over time, we can add, miracles and disasters).

The Model

2 Aggregate Demand and Supply

- Production Function:

Consider a pure private economy with two inputs: L and K . The production function is given by:

$$Y = F(K, L, t) \quad (1)$$

- Consumption and Saving

Output is divided into consumption and saving i.e.,

$$Y = C + I$$

Solow assumed that households save a fixed percent of their income. In this economy, saving immediately translates into investment. The consumption and saving functions are given by:

$$\begin{aligned} C &= (1 - s)Y ; (1 - s) = \frac{c}{y} \\ \Rightarrow S &= sY = sF(K, L) ; 0 \leq s \leq 1 \\ \Rightarrow S &= sF(K, L, t) = I \end{aligned} \quad (2)$$

⁰I am truly grateful to my students who helped me prepare this handout. This handout is heavily based on Macroeconomics textbooks written by Campante et al. and Sala-i- Martin and, therefore, does not reflect my original scholarship.

- Capital and Labor:

- Capital accumulation follows the following law of motion.

$$\begin{aligned}\dot{K} &= I - \delta K, \delta > 0 \\ \Rightarrow \dot{K} &= sF(K, L, t) - \delta K\end{aligned}\quad (3)$$

- Population (=Labor Force) grows at a constant rate n i.e.,

$$L_t = L_0 e^{nt} = e^{nt}, \quad L_0 = 1 \quad (4)$$

We assume that technology is fixed for the time being, allowing us to drop t from the production function.

3 The Neoclassical Production Function:

The production function we are going to use in this model is Neoclassical. It is very important that you understand these properties.

$Y = F(K, L)$ satisfies the following three important Neoclassical properties:

1. MP_K and MP_L are positive and diminishing i.e.,

$$\begin{aligned}\frac{\partial F}{\partial K} &> 0, \quad \frac{\partial F}{\partial L} > 0 \\ \frac{\partial^2 F}{\partial K^2} &< 0, \quad \frac{\partial^2 F}{\partial L^2} < 0\end{aligned}$$

2. Constant Returns to Scale (CRS):

$$F(\lambda K, \lambda L) = \lambda \cdot F(K, L), \quad \forall \lambda > 0$$

3. Inada Conditions:

$$\begin{aligned}\lim_{K \rightarrow \infty} \frac{\partial F}{\partial K} &= \lim_{L \rightarrow \infty} \frac{\partial F}{\partial L} = 0 \\ \text{and } \lim_{K \rightarrow 0} \frac{\partial F}{\partial K} &= \lim_{L \rightarrow 0} \frac{\partial F}{\partial L} = \infty\end{aligned}$$

Note: If a production function satisfies these three conditions, we call it Neoclassical Production Function (See Romer 2018, for detail).

Production Function in intensive form:

CRS allows us to write the production function in intensive/per capita form:

$$\begin{aligned}Y &= F(K, L) \\ \Rightarrow \frac{Y}{L} &= F\left(\frac{K}{L}, \frac{L}{L}\right) \\ \Rightarrow y &= F(k, 1) \\ \Rightarrow y &= f(k); \quad k \equiv \frac{K}{L}, \quad y \equiv \frac{Y}{L}\end{aligned}$$

Example of Neoclassical Production Function :

Cobb-Douglas production function is an classic example of Neoclassical production function:

$$Y = AK^\alpha L^{1-\alpha}$$

$$(1) \quad \frac{\partial Y}{\partial K} = \text{MP}_K = \alpha AK^{\alpha-1} L^{1-\alpha} \\ = \alpha Ak^{\alpha-1} > 0$$

$$\frac{\partial^2 Y}{\partial K^2} = \alpha(\alpha-1)AK^{\alpha-2}L^{1-\alpha}$$

Since $(\alpha-1) < 0$, $\frac{\partial^2 Y}{\partial K^2} < 0$, MP_K diminishing

$$\frac{\partial Y}{\partial L} = \text{MP}_L = (1-\alpha)AK^\alpha L^{-\alpha} \\ = (1-\alpha)Ak^\alpha > 0$$

$$\frac{\partial^2 Y}{\partial L^2} = -\alpha(1-\alpha)AK^\alpha L^{-(\alpha+1)} < 0$$

(2) CRS:

$$Y = AK^\alpha L^{1-\alpha}$$

$$\text{Now, } \lambda Y = A(\lambda K)^\alpha (\lambda L)^{1-\alpha} \\ = A\lambda^\alpha K^\alpha \cdot \lambda^{1-\alpha} \cdot L^{1-\alpha} \\ = A\lambda^{\alpha+1-\alpha} \cdot K^\alpha L^{1-\alpha} \\ = A \cdot \lambda \cdot K^\alpha L^{1-\alpha} \\ = \lambda \cdot (AK^\alpha L^{1-\alpha}) \\ = \lambda \cdot Y$$

(3) Inada Conditions:

$$\lim_{K \rightarrow \infty} \text{MP}_K = \lim_{K \rightarrow \infty} \alpha AK^{\alpha-1} L^{1-\alpha} \\ = \lim_{K \rightarrow \infty} \alpha \cdot A \cdot K^{-(1-\alpha)} L^{1-\alpha} \\ = \lim_{K \rightarrow \infty} \frac{\alpha AL^{1-\alpha}}{K^{1-\alpha}} \\ = \frac{\alpha AL^{1-\alpha}}{\infty} \\ = 0$$

Similarly,

$$\lim_{L \rightarrow \infty} \text{MP}_L = \lim_{L \rightarrow \infty} (1-\alpha)AK^\alpha L^{-\alpha} \\ = \lim_{L \rightarrow \infty} \frac{(1-\alpha)AK^\alpha}{L^\alpha} \\ = \frac{(1-\alpha)AK^\alpha}{\infty} \\ = 0$$

And,

$$\lim_{K \rightarrow 0} \text{MP}_K = \lim_{K \rightarrow 0} \frac{\alpha AL^{1-\alpha}}{K^{1-\alpha}} = \frac{\alpha AL^{1-\alpha}}{0} = \infty \\ \lim_{L \rightarrow 0} \text{MP}_L = \lim_{L \rightarrow 0} \frac{(1-\alpha)AK^\alpha}{L^\alpha} = \infty$$

So, Cobb-Douglas production function follows Neoclassical properties.
Intensive form:

With CRS,

$$\begin{aligned}
 Y &= AK^\alpha L^{1-\alpha} \\
 \Rightarrow \frac{Y}{L} &= AK^\alpha \frac{L^{1-\alpha}}{L} \\
 \Rightarrow y &= AK^\alpha L^{-\alpha} \\
 \Rightarrow y &= A\left(\frac{K}{L}\right)^\alpha = Ak^\alpha
 \end{aligned} \tag{5}$$

4 Fundamental Equation of the Solow Model

From the law of motion of capital,

$$\begin{aligned}
 \dot{K} &= sY - \delta K \\
 \Rightarrow \frac{\dot{K}}{L} &= s\frac{Y}{L} - \delta\frac{K}{L} \\
 \Rightarrow \frac{\dot{K}}{L} &= sf(k) - \delta k
 \end{aligned} \tag{6}$$

R.H.S is in per capita form but the L.H.S is not.
We know,

$$\begin{aligned}
 k &= \frac{K}{L} \\
 \Rightarrow \log k_t &= \log K_t - \log L_t \\
 \Rightarrow \frac{d \log k_t}{dt} &= \frac{d \log K_t}{dt} - \frac{d \log L_t}{dt} \\
 \Rightarrow \frac{\dot{k}}{k} &= \frac{\dot{K}}{K} - \frac{\dot{L}}{L} \\
 \Rightarrow \frac{\dot{k}}{k} &= \frac{\dot{K}}{K} - n \text{ (Since, the labor force grows at } n \text{ rate)} \\
 \Rightarrow \dot{k} &= \frac{\dot{K}}{K} \times \frac{K}{L} - nk \\
 \Rightarrow \dot{k} &= \frac{\dot{K}}{L} - nk
 \end{aligned} \tag{7}$$

Using (7) in (6),

$$\begin{aligned}
 \dot{k} + nk &= sf(k) - \delta k \\
 \Rightarrow \dot{k} &= sf(k) - \delta k - nk \\
 \Rightarrow \dot{k} &= sf(k) - (\delta + n)k
 \end{aligned} \tag{8}$$

(8) gives the fundamental equation of the Solow model where $(\delta + n)$ is the effective depreciation rate for $k = \frac{K}{L}$. Capital per person reduces because capital wears out and population is growing.

5 Balanced Growth Path

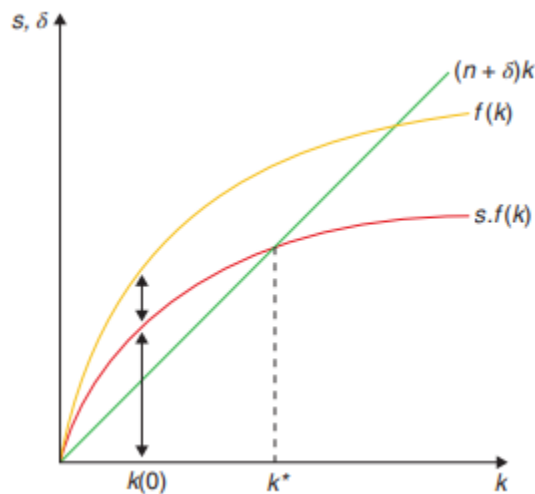
A balanced growth path is a situation in which various quantities grow at constant rates. In the Solow Model, BGP corresponds to steady state equilibrium where $\dot{k} = 0$.

Denote $k^* \equiv$ steady state level of k .

Now setting $\dot{k} = 0$ in (8),

$$\begin{aligned} 0 &= sf(k) - (n + \delta)k \\ \Rightarrow sf(k^*) &= (n + \delta)k^* \end{aligned} \quad (9)$$

(9) gives the condition for steady state (when capital stops). The situation is shown in the graph:



5.1 Growth rate of variables:

Since k stops in the BGP, $y = f(k)$ also stops i.e.,

$$\gamma_k = 0 = \gamma_y$$

$$\text{Since, } c^* = (1 - s)f(k^*), \quad g_k = 0 \Rightarrow \gamma_c = 0$$

Hence, $\gamma_k = \gamma_c = \gamma_y = 0$ in the BGP.

Per capita variables do not grow in the long run. That is, it is a growth model without growth.

However, aggregate variables grow at positive rates in the BGP:

$$\begin{aligned} y &= \frac{Y}{L} \\ \Rightarrow \log y &= \log Y - \log L \\ \Rightarrow \frac{d \log y}{dt} &= \frac{d \log Y}{dt} - \frac{d \log L}{dt} \\ \Rightarrow \gamma_y &= \gamma_Y - \gamma_L \\ \Rightarrow 0 &= \gamma_Y - n \\ \Rightarrow \gamma_Y &= n \end{aligned}$$

That is, in the BGP, aggregate output Y grows at a constant rate, n . Similarly,

$$\begin{aligned} k &= \frac{K}{L} \\ \Rightarrow \gamma_k &= \gamma_K - \gamma_L \\ \Rightarrow 0 &= \gamma_K - n \\ \Rightarrow \gamma_K &= n \end{aligned}$$

and

$$\begin{aligned} c &= \frac{C}{L} \\ \Rightarrow \gamma_c &= \gamma_C - \gamma_L \\ \Rightarrow \gamma_C &= n \end{aligned}$$

In the Solow model, without technological progress, per capita variables c, k and y grow at the constant rate 0, but aggregate variables grow by the growth rate of population n .

5.2 Solving for k^* , y^* and c^* :

Using Cobb-Douglas production function, we can get analytical expressions (closed-form solution) for k^*, y^* and c^* .

At the steady state,

$$\begin{aligned} sf(k^*) &= (n + \delta)k^* \\ \text{Now, with } Y &= AK^\alpha L^{1-\alpha} \\ \Rightarrow \frac{Y}{L} &= AK^\alpha \frac{L^{1-\alpha}}{L} \\ \Rightarrow y &= Ak^\alpha = f(k) \end{aligned}$$

$\therefore$ Steady state condition becomes:

$$\begin{aligned} s \cdot Ak^{*\alpha} &= (n + \delta)k^* \\ \Rightarrow sA &= \frac{n + \delta}{k^{*\alpha}} \cdot k^* \\ \Rightarrow k^{*1-\alpha} &= \frac{sA}{n + \delta} \\ \Rightarrow k^* &= \left(\frac{sA}{n + \delta} \right)^{\frac{1}{1-\alpha}}, \text{ constant.} \end{aligned}$$

k^* rises with s and A , but falls with n and δ .

Output per capita:

$$\begin{aligned} \text{Now, } y^* &= Ak^\alpha = A \cdot \left(\frac{sA}{n + \delta} \right)^{\frac{\alpha}{1-\alpha}} \\ &= A \cdot A^{\frac{\alpha}{1-\alpha}} \cdot \left(\frac{s}{n + \delta} \right)^{\frac{\alpha}{1-\alpha}} \\ &= A^{1+\frac{\alpha}{1-\alpha}} \left(\frac{s}{n + \delta} \right)^{\frac{\alpha}{1-\alpha}} \\ \Rightarrow y^* &= A^{\frac{1}{1-\alpha}} \cdot \left(\frac{s}{n + \delta} \right)^{\frac{\alpha}{1-\alpha}}, \text{ constant.} \end{aligned}$$

Consumption per capita:

$$\begin{aligned} c^* &= (1 - s)y^* \\ \Rightarrow c^* &= (1 - s)A^{\frac{1}{1-\alpha}} \left(\frac{s}{n + \delta} \right)^{\frac{\alpha}{1-\alpha}}, \text{ constant.} \end{aligned}$$

6 Transitional Dynamics

The Solow model generates growth in the transition to the BGP. Recall the fundamental equation of the model.

$$\dot{k} = sf(k) - (n + \delta)k$$

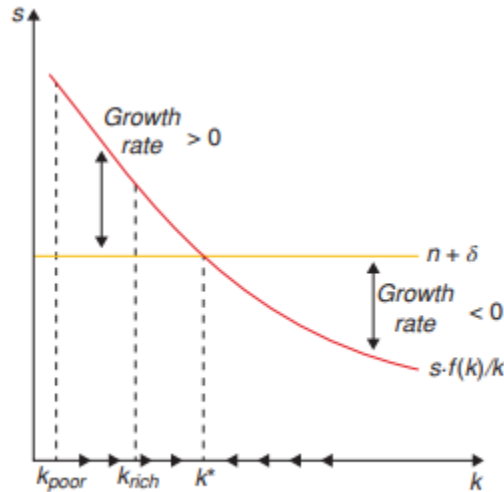
Divide this by k ,

$$\frac{\dot{k}}{k} = \gamma_k = \frac{sf(k)}{k} - (n + \delta) \quad (1)$$

(1) says that γ_k is given by the difference between two terms: $\frac{sf(k)}{k}$ and $(n + \delta)$. Note that

- $\frac{sf(k)}{k}$ curve is downward sloping, and note that $\lim_{k \rightarrow \infty} \frac{sf(k)}{k} = 0$ and $\lim_{k \rightarrow 0} \frac{sf(k)}{k} = \infty$.
- $(n + \delta)$ is a horizontal line crossing the vertical axis at $(n + \delta)$.

The vertical distance between these two curves equals the growth rate of k , and the point of intersection corresponds to BGP. Note that this intersection occurs only once; hence, solution exists and it is unique. Figure 2 shows this situation.



Now output,

$$\begin{aligned}
 y &= Ak^\alpha \\
 \Rightarrow \frac{dy}{dt} &= \alpha Ak^{\alpha-1} \cdot \frac{dk}{dt} = \alpha \frac{Ak^\alpha}{k} \cdot \dot{k} \\
 \Rightarrow \frac{\dot{y}}{y} &= \frac{\alpha Ak^\alpha \cdot \dot{k}}{k \cdot y} = \alpha \frac{\dot{k}}{k} \\
 \Rightarrow \frac{\dot{y}}{y} &= \gamma_y = \alpha \cdot \gamma_k
 \end{aligned}$$

This gives that growth rate of output per capita in the transition to the BGP (transitional growth).

Formal treatment:

Note that with $(n + \delta)k$ line having a constant slope, we require $\dot{k} < 0$ if $k > k^*$ and $\dot{k} > 0$ if $k < k^*$. This requires the slope of $sf(k)$ must decline and eventually cut the $(n + \delta)k$ line.

Mathematically, differentiating the $\dot{k}$ with respect to capital,

$$\frac{d\dot{k}}{dk} = sf'(k) - (n + \delta) \quad (2)$$

Since we are interested about BGP, we want to study the dynamics of k around BGP k^* . Then evaluating (2) by k^* ,

$$\left. \frac{d\dot{k}}{dk} \right|_{k=k^*} = sf'(k^*) - (n + \delta)$$

We see that BGP requires $\left. \frac{d\dot{k}}{dk} \right|_{k=k^*} < 0$. That is, the difference between the slope of the saving function and the slope of the depreciation line must be negative.

In the case of Cobb-Douglas case this requires,

$$\begin{aligned}
 \dot{k} &= sAk^\alpha - (n + \delta)k \\
 \Rightarrow \frac{d\dot{k}}{dk} &= s \cdot A\alpha k^{\alpha-1} - (n + \delta) \\
 &= sA\alpha \left[\left(\frac{sA}{n + \delta} \right)^{\frac{1}{1-\alpha}} \right]^{\alpha-1} - (n + \delta) \\
 &= sA\alpha \left(\frac{sA}{n + \delta} \right)^{-1} - (n + \delta) \\
 &= \alpha(n + \delta) - (n + \delta) \\
 &= (n + \delta)(\alpha - 1) \\
 \therefore \frac{d\dot{k}}{dk} < 0 &\Rightarrow (n + \delta)(\alpha - 1) < 0 \\
 &\Rightarrow \alpha < 1
 \end{aligned}$$

$\therefore$ BGP requires $\alpha < 1$ i.e., diminishing returns to k . You should memorize this result by heart.

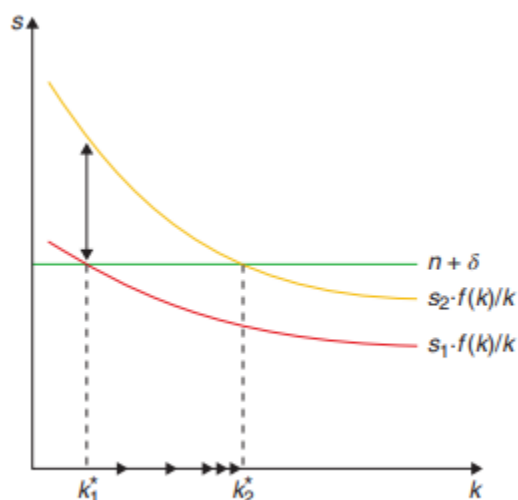
With diminishing returns, when k is low, $MPK = f'(k)$ is relatively high. Since s is constant, by assumption, $sf'(k)$ is also relatively high. Again, Since k depreciates at $(n + \delta)$ rate, a constant, $\dot{k}$ is also relatively high. In fact, for $k < k^*$, $\dot{k} > 0$. Conversely, for $k > k^*$, $\dot{k} < 0$.

7 Policy Experiments:

Suppose that the economy's initial $k^* = k_1^*$. We can now change different parameters to study their impacts on growth and level of the variables of the model.

7.1 Impact of saving:

A permanent increase in saving from s_1 to s_2 shifts $sf(k)$ schedule to the right. Hence, the intersection of $sf(k)$ and $(n + \delta)$ line also shifts to the right, from k_1^* to k_2^* . An increase in saving temporarily generates growth (transitional growth). But in the BGP, y , c and k growth stops once again at k_2^* .



A permanent increase in technology also has similar effect.

7.2 Dynamic inefficiency:

Clearly, steady state/BGP k^* is depending on s . For each s , we have a corresponding k^* .

So, define $k^* = k^*(s)$, $\frac{dk^*(s)}{ds} > 0$

$$\begin{aligned}\text{Then } c^* &= (1-s)f(k^*(s)) \\ \Rightarrow c^* &= f(k^*(s)) - sf(k^*(s)) \\ \Rightarrow c^* &= f(k^*(s)) - (n+\delta)k(s)\end{aligned}$$

Consumer's welfare is maximized with the maximum per capita consumption and is given by:

$$\begin{aligned}\frac{dc^*}{ds} &= f'(k)\frac{dk}{ds} - (n+\delta)\frac{dk}{ds} \\ &= [f'(k) - (n+\delta)]\frac{dk}{ds} = 0\end{aligned}$$

Since $\frac{dk}{ds} > 0$, $f'(k) - (n+\delta) = 0$

$$\Rightarrow \boxed{f'(k) = (n+\delta)}$$

Clearly c is maximized when $f'(k) = n + \delta$. Let's denote the level of capital which satisfies this condition by k_g . Then the corresponding consumption:

$c_g = f(k_g) - (n+\delta)k_g$, is called *the golden rule of consumption rate*.

If $s > s_g$, then it is possible to increase consumption on BGP and also in the transition to the BGP. We refer to this situation, when everyone would be made better off by an alternative allocation, as one of dynamic inefficiency. The Solow Model leaves some scope for this type of dynamic inefficiency because saving ratio was imposed on the consumer/household.

8 Absolute and Conditional Convergence

Recall that

$$\frac{\partial \gamma_k}{\partial k} = \frac{s}{k}(f'(k) - \frac{f(k)}{k}) < 0$$

This implies that a country with lower capital stock grows faster. Is there any convergence across economies?

The Solow model suggest that economies that are structurally similar in the sense that they have the same values of the parameters s, n , and δ and also have the same production function, $F(\cdot)$, have the same BGP values k^*, y^* , no matter what their initial conditions are. This hypothesis is known as conditional convergence. This hypothesis match the data (e.g., European Countries).

Note that the model does not say that there would be absolute convergence. The model does not say all countries will converge to the same BGP equilibrium, regardless of the fundamentals.

8.1 Cross country differences in income (levels)

We have seen that $\gamma_k = \gamma_y = \gamma_c = 0$ in the Solow Model (within a country). And like countries converge to like countries (conditional convergence). But can it help us understand income differential?

Today (2020) income varies \$138,000 in Qatar to \$80,000 in Norway to \$700 in Burundi. Can the Solow Model answer the question:

Why are the poor countries poor? Why is Qatar 200 times richer than Burundi?

To answer this question, recall that along the BGP,

$$y^* = A^{\frac{1}{1-\alpha}} \left(\frac{s}{n+\delta} \right)^{\frac{\alpha}{1-\alpha}}$$

(i). For simplicity, assuming $A = 1$ and $n = 0$, then

$$y^* = \left(\frac{s}{\delta} \right)^{\frac{\alpha}{1-\alpha}}$$

Consider two countries: Country 1 and 2, then

$$\frac{y_1}{y_2} = \frac{\left(\frac{s_1}{\delta} \right)^{\frac{\alpha}{1-\alpha}}}{\left(\frac{s_2}{\delta} \right)^{\frac{\alpha}{1-\alpha}}} = \left(\frac{s_1}{s_2} \right)^{\frac{\alpha}{1-\alpha}}$$

Then cross country differences in income come from differences in saving and capital's share (α) across countries. The following table shows income differences for various values of s and α .

	$\left(\frac{y_1}{y_2} - 1 \right)$		
$\frac{s_1}{s_2}$	$\alpha = \frac{1}{3}$	$\alpha = \frac{1}{2}$	$\alpha = \frac{2}{3}$
1	0	0	0
1.5	22%	50%	125%
2	41%	100%	300%
3	73%	200%	800%

Even with $\frac{s_1}{s_2} = 3$ and $\alpha = \frac{2}{3}$, the first country is 800% richer than the 2nd country. But $\frac{138000}{700} \times 100 = 19714 \approx 20,000$, shows Qatar is 20,000% richer than Burundi. So, we need to look for other reasons.

(ii). Differences in income may come from differences in the total factor productivity, A .

Q. How large do these differences need to be to explain output differential?

Let $\alpha = \frac{2}{3}$, (after Presscott, Parente 2002). Then $A^{\frac{1}{1-\alpha}} = A^3$. Now let s, δ , and n are the same across countries and just focus on only A .

$$\frac{y_1}{y_2} = \frac{A_1^{\frac{1}{1-\alpha}}}{A_2^{\frac{1}{1-\alpha}}} = \left(\frac{A_1}{A_2} \right)^{\frac{1}{1-\alpha}}$$

$$\alpha = \frac{2}{3}, \frac{y_1}{y_2} = \left(\frac{A_1}{A_2} \right)^3$$

$$\text{Now if } \frac{A_1}{A_2} = 3, \text{ then } \frac{y_1}{y_2} = 27$$

Parente and Presscott (2002) used this to estimate how much the productivity gap should there be to account for observed gap in

income relative to USA. The following shows this for some of the countries:

Country	Relative Income	Relative TFP
UK	60%	80%
Columbia	22%	64%
Paraguay	16%	59%
Pakistan	10%	51%

Solow Model requires substantial gap in factor productivity to explain existing cross country differences in income.

This begs the question of what makes the productivity so different across countries?

To answer this question we need to study New Growth Theory. For now, we focus on growth itself.

9 The Solow Model with exogenous technological change

Let $\frac{\dot{A}}{A} = \gamma_A = g$, where g is a constant. That is, technology grows by an exogenous g rate.

Then,

$$\begin{aligned} y &= Ak^\alpha \\ \Rightarrow \log y &= \log A(t) + \alpha \log k(t) \\ \Rightarrow \frac{d \log y}{dt} = \gamma_y &= \frac{d \log A}{dt} + \alpha \frac{d \log k(t)}{dt} \\ \therefore \gamma_y &= \gamma_A + \alpha \gamma_k \end{aligned}$$

Since along BGP k is constant, $\gamma_y = \gamma_A = g$.

In the long run, technological growth is the only source of growth in the per capita output.

How can we introduce this technological change in the model?

It tuns out only labor augmenting (Harrod-neutral) technological progress is consistent with the BGP. So, we attach technology to the the labor force.

Define

$$\begin{aligned} E &\equiv L_t \cdot e^{gt} \\ &= L_0 e^{nt} \cdot e^{gt} \\ &= L_0 e^{(n+g)t} \\ \text{Assume } L_0 = 1 &\Rightarrow E = e^{(n+g)t} \end{aligned}$$

Production Function:

$$\begin{aligned} Y = F(K_t, E_t) &\Rightarrow \frac{Y}{E} = F\left(\frac{K}{E}, 1\right) \\ &\Rightarrow \hat{y} = f(\hat{k}) \end{aligned}$$

Define $\hat{k} \equiv \frac{K}{E}$

So,

$$\begin{aligned}
 \log \hat{k} &= \log K - \log E \\
 \Rightarrow \frac{d \log \hat{k}}{dt} &= \frac{d \log \hat{K}}{dt} - \frac{d \log E}{dt} \\
 \Rightarrow \frac{\dot{\hat{k}}}{\hat{k}} &= \gamma_{\hat{k}} = \frac{\dot{K}}{K} - \frac{\dot{E}}{E} \\
 \Rightarrow \frac{\dot{\hat{k}}}{\hat{k}} &= \frac{I - \delta K}{K} - (n + g) \\
 \Rightarrow \frac{\dot{\hat{k}}}{\hat{k}} &= \frac{sY}{K} - (n + g + \delta) \\
 \Rightarrow \frac{\dot{\hat{k}}}{\hat{k}} &= \frac{s \frac{Y}{E}}{\frac{K}{E}} - (n + g + \delta) \\
 \Rightarrow \frac{\dot{\hat{k}}}{\hat{k}} &= \frac{s\hat{y}}{\hat{k}} - (n + g + \delta) \\
 \Rightarrow \frac{\dot{\hat{k}}}{\hat{k}} &= \frac{sf(\hat{k})}{\hat{k}} - (n + g + \delta)
 \end{aligned}$$

$$\text{Along BGP, } \frac{\dot{\hat{k}}}{\hat{k}} = \gamma_{\hat{k}} = 0 \Rightarrow \frac{sf(\hat{k})}{\hat{k}} - (n + g + \delta) = 0$$

$$\boxed{\Rightarrow sf(\hat{k}) = (n + g + \delta)\hat{k}} \rightarrow \text{(Condition for BGP)}$$

9.1 Growth rates

In the steady state, $\dot{\hat{k}} = 0$. Therefore,

$$\begin{aligned}
 \frac{\dot{K}}{K} - \frac{\dot{E}}{E} &= 0 \\
 \Rightarrow \frac{\dot{K}}{K} &= g_K = (n + g) \\
 \text{Now, } \hat{k} = \frac{K}{E} &= \frac{K}{L \cdot e^{gt}} = k \cdot e^{-gt} \\
 \therefore \log \hat{k} &= \log k - gt \\
 \Rightarrow \frac{d \log \hat{k}}{dt} &= \frac{d \log k}{dt} - g \\
 \Rightarrow \frac{\dot{\hat{k}}}{\hat{k}} &= \frac{\dot{k}}{k} - g \\
 \Rightarrow 0 &= \frac{\dot{k}}{k} - g \\
 \Rightarrow \boxed{\frac{\dot{k}}{k} = g = \gamma_k}
 \end{aligned}$$

Similarly, $\gamma_k = \gamma_y = \gamma_c = g$ (find it)

Since $\hat{y} = f(\hat{k})$, $\gamma_{\hat{k}} = 0 \Rightarrow \gamma_{\hat{y}} = 0$

$$\begin{aligned}
 \text{But } \hat{y} &= \frac{Y}{E} = \frac{Y}{L \cdot e^{gt}} \\
 \Rightarrow \hat{y} &= \frac{Y}{L} \cdot e^{-gt} \\
 \Rightarrow \boxed{\gamma_y = g}
 \end{aligned}$$

Similarly, $\hat{c} = (1 - s)f(\hat{k}) \Rightarrow \boxed{\gamma_c = g}$

$$\therefore \gamma_k = \gamma_y = \gamma_c = g$$

And one can show K , L , and C grow by the rate $(n+g)$. Although $\hat{k}$ stops in the BGP, k grows by the rate of technological change. This is what we need to confirm stylized fact iii.

10 Conclusion (From Campante et al., 2021)

The Solow model shows that capital accumulation by itself cannot sustain growth in per capita income in the long run. This is because accumulation runs into diminishing marginal returns. At some point the capital stock becomes large enough – and its marginal product correspondingly small enough – that a given savings rate can only provide just enough new capital to replenish ongoing depreciation and increases in labour force.

Alternatively, if we introduce exogenous technological change that increases productivity, we can generate long-run growth in income per capita, but we do not really explain it. In fact, any differences in long-term growth rates come from exogenous differences in the rate of technological change – we are not explaining those differences, we are just assuming them! As a result, nothing within the model tells you what policy can do about growth in the long run. That said, we do learn a lot about growth in the transition to the long run, about differences in income levels, and how policy can affect those things.

There are, however, clear lessons about: (i) convergence – the model predicts conditional convergence; (ii) dynamic inefficiency – it is possible to save too much in this model; and (iii) long-run differences in income – they seem to have a lot to do with differences in productivity. Very importantly, the model also points at the directions we can take to try and understand long-term growth. We can have a better model of savings behaviour: how do we know that individuals will save what the model says they will save? And, how does that relate to the issue of dynamic inefficiency?

We can look at different assumptions about technology: maybe we can escape the shackles of diminishing returns to accumulation? Or can we think more carefully about how technological progress comes about? The New Growth Theories will answer many of these questions.

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